

Scenario 1: Physical Robot Deployment (BonicBot + ROS 2)

Robotics & Edge AI Workshop — Architecture, Theory, and Step-by-Step Execution Guide

1. System Architecture & Theory

Core Concept: In a networked robotics architecture, physical sensors (camera, LiDAR, IMU) and motor drivers run on a dedicated onboard computer (e.g., Raspberry Pi 4/5 or NVIDIA Jetson) running **Ubuntu Linux and ROS 2**. The robot acts as an edge node, and remote workstations interact with it via a lightweight bridge protocol.

Key Subsystems:

- **Edge Sensing & YOLO:** The camera feed is ingested into a ROS 2 camera node. YOLO runs directly on the robot hardware, predicting bounding boxes in normalized coordinates [cx, cy, w, h] to minimize data payload.
- **ROSBridge WebSocket Server:** Exposes ROS 2 topics and services via JSON over WebSockets (default Port 9090).
- **Client Bridge (Laptop):** Uses bonicbot-bridge (built on roslibpy) to subscribe to camera frames and YOLO detection telemetry, allowing remote telemetry display and control.

Layer	Component	Role / Protocol
Hardware	Robot Camera & Sensors	Captures raw video frames at source
Robot OS	ROS 2 + YOLO Pipeline	Runs local object detection; extracts [cx, cy, w, h]
Transport	rosbridge_server (Port 9090)	Serializes ROS topics to JSON WebSockets over Wi-Fi
Client / PC	bonicbot-bridge + OpenCV	Receives JSON telemetry, renders bounding boxes & UI

2. Prerequisites & Setup

- **Laptop & Robot must be on the same Wi-Fi network.**
- Client Python packages installed on laptop: `pip install bonicbot-bridge opencv-python numpy`.

3. Step-by-Step Commands

Step A: On the Robot (SSH / Terminal)

```
# 1. Find robot IP address
hostname -I

# 2. Launch the robot system with live camera
ros2 launch my_bot robot_system.launch.py use_sim_time:=false use_real_camera:=True

# 3. Ensure ROSBridge WebSocket server is active (Port 9090)
ros2 launch rosbridge_server rosbridge_websocket_launch.xml
```

Step B: On Laptop (Windows PowerShell Verification)

```
# 1. Ping the robot to test IP connectivity
ping

# 2. Verify Port 9090 (WebSocket) is reachable
Test-NetConnection -ComputerName -Port 9090
```

Step C: Run Client Script on Laptop

```
# In bonicbot_vision.py, set ROBOT_IP = ''  
python bonicbot_vision.py
```

4. Workshop Troubleshooting & FAQ

- **'RosTimeoutError: Failed to connect to ROS':** Robot IP is incorrect, robot is off, or Port 9090 is blocked by firewall.
- **AP / Wi-Fi Isolation:** Some university/corporate Wi-Fi blocks device-to-device communication. Use a mobile hotspot if ping fails.
- **Lag / High Latency:** Reduce camera resolution in ROS launch file to 320x240 or 640x480.